Venkata Lakshmi Parimala Pasupuleti

Software Engineer, Agents · Agentic Systems · RAG & Context Engineering · LLM Evaluation · Multi-Agent Architectures · Production AI · Python · LangGraph

venkatalakshmiparimalap@gmail.com | +1-703-261-5410 | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | AgentProbe (eval toolkit) | Virginia · Open to Onsite NYC

PROFESSIONAL SUMMARY

Software engineer with 9+ years of experience (7+ in production), specializing in **agentic systems, multi-agent architectures, RAG pipelines, context engineering, and evaluation frameworks**. Built two production agentic systems: **Leasel** — an 8-agent LangGraph StateGraph with multi-step planning, confidence-gated HITL escalation, and a 1,240-case eval harness (93.6% precision, 87.9% F1); **AgentProbe** — open-source LLM evaluation toolkit with a 5-metric LLM-as-judge eval engine, CI-gated regression detection, and dynamic **tool selection** across 5 providers. Currently at Intuit, owning end-to-end: production **agentic pipeline** integrating Anthropic Claude into financial apps — **context engineering**, **token optimization**, RAG grounding, eval harness catching 3 prompt regressions before release, zero hallucination incidents post-deploy. **Agentic infrastructure** discipline: designed state machines before code, calibrated LLM-as-judge prompts, integrated **experimentation** loops into every eval cycle. Fine-tuning background: HuggingFace + PyTorch classification model (GMU, presented to faculty). M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026). — github.com/venkatalakshmiparimala/AgentProbe

CORE COMPETENCIES

Agentic Systems & Pipelines | Multi-Agent Architectures | Context Engineering | Token Optimization | RAG Systems & Hybrid Retrieval | LLM Evaluation Frameworks | LLM-as-Judge & Eval CI | Tool Selection & LLM Routing | Multi-Step Planning | Agentic Infrastructure | SOTA Model Integration | Generative AI & Production LLMs

EXPERIENCE

Intuit

Jun 2026 – Present · Virginia, USA

Software Engineer — Production Agentic AI · Context Engineering · LLM Evaluation · Python · AWS

- Own **production agentic pipeline** integrating Anthropic Claude into financial apps end-to-end: discover use cases, build with **context engineering** and RAG grounding, productionize with guardrails, evals, monitoring, and fallback paths; every Claude response grounded in retrieved customer financial records for full citation traceability.
- Apply **token optimization** and prompt management: structured-output schemas constrain model outputs for downstream validation; prompt versioning enables regression tracking; **context gathering** layer retrieves only relevant records, keeping context windows efficient and reducing cost by design.
- Build **evaluation framework**: LLM-as-judge scoring with golden datasets and CI regression detection; caught 3 prompt regressions before release; zero hallucination incidents post-deployment; add **experimentation** loops for prompt A/B testing and model comparison against **SOTA models**; implement monitoring via OpenTelemetry for full production AI visibility.
- Build guardrails: input validation, sensitive-data redaction, output schema enforcement, fallback paths; containerize services with Docker and ship through GitHub Actions to AWS with logging for production observability.

Entrata (Apartment & Affordable Housing SaaS)

Oct 2020 – Jul 2024 · 100K+ MAU, 1M+ property records

Senior Software Engineer — Python · Java · AWS · PostgreSQL · Distributed Systems

- Shipped 20+ production features end-to-end (Python, Java/Spring Boot, PHP, PostgreSQL, React) across 120+ Agile sprints; cross-functional delivery with Legal, Compliance, Operations, and Product; reviewed 300+ PRs; mentored junior engineers.
- Built automation tools in Python/Bash eliminating 80% of manual operational work; built internal debugging and data investigation tools cutting triage time 40%; reduced 200+ SQL queries' runtime by 30% via execution plan analysis and indexing.
- Owned HUD income-limit ETL pipeline for 7 years: federal source rows → 180K domain records with validation, rollback checks, and compliance enforcement; zero violations across 7 annual release cycles.
- AWS S3 PHI migration with Libsodium encryption (100% PHI compliance, zero security incidents); Docker containerization cutting microservices execution time 50%; RabbitMQ async processing with idempotency guarantees for bulk tenant record updates.

Entrata India (Db Xento Systems)

Sep 2018 – Oct 2020 · Pune, India

Junior Software Engineer — Python · PHP · Java · PostgreSQL · AWS

- Built property management modules and compliance data pipelines processing 1.6M HUD/Tax Credit records; led AWS S3 cloud migration; created 56 state-specific housing certification documents.

George Mason University — Research Associate / Teaching Assistant

Aug 2024 – May 2026

AI/ML Research · Fine-Tuning · PyTorch · HuggingFace · Statistical Modeling

- Fine-tuned classification model with HuggingFace Transformers + PyTorch: full training loop (data prep → fine-tuning → eval → deploy on Quart), presented to GMU faculty panel. Built predictive models ($R^2 = 0.977$, 190+ countries); applied Isolation Forest anomaly detection. Mentored 600+ students.

PORTFOLIO PROJECTS

AgentProbe — Multi-Agent Evaluation Infrastructure & LLM Eval Toolkit

Agentic Systems · Eval Frameworks · Tool Selection

2025 · Python, FastAPI (15 endpoints), LangGraph (3 agents), ChromaDB, Anthropic, OpenAI, Azure OpenAI, Groq, Ollama · github.com/venkatalakshmiparimala/AgentProbe

- **5-metric LLM-as-judge evaluation framework** (factual accuracy, hallucination rate, retrieval quality, completeness, compliance); CI-integrated regression detection blocks production deploys when quality drops below threshold; **context engineering**: judge prompts designed, tested, and calibrated independently before gating production runs.
- 3-agent LangGraph **agentic infrastructure** (Orchestrator, RAG Tracer, Auto Fix) with multi-step planning and inter-agent state handoffs; LLM Router for dynamic **tool selection** and task-based routing across 5 **SOTA model** providers (Anthropic, OpenAI, Azure OpenAI, Groq, Ollama); 15-endpoint FastAPI backend; 100% local, zero vendor lock-in; used to run **experimentation** across providers for Leasel eval (1,240 test cases).

Leasel — 8-Agent Agentic Platform with RAG & Multi-Step Planning

Multi-Agent · LangGraph · Eval Harness

2026 · Python, LangGraph (8-agent StateGraph), FastAPI, TF-IDF RAG (hybrid + reranking), SQLite, Anthropic/OpenAI APIs · github.com/venkatalakshmiparimala/Lease_complaince_rag

- **8-agent LangGraph StateGraph** with real conditional edges and **multi-step planning**: triage → retrieval → Config/History/StructuredData parallel agents → reasoning → validation → confidence-gated HITL escalation; correctly traces false-ineligibility ticket to misconfigured setting tagged to specific release — the canonical complex **agentic pipeline** test case.
- **RAG system** with hand-rolled TF-IDF + cosine reranking + section-aware chunking: **context engineering** built into retrieval (returns exact relevant clauses, not raw document dumps); **token optimization** by design — context windows stay efficient; 100% citation correctness, 0% unsupported-claim rate.
- **Evaluation framework** via AgentProbe LLM-as-judge CI harness: 93.6% precision, 87.9% F1 across 1,240 test cases; **agentic infrastructure** enforced in code: LLM reasons, human or deterministic validator decides every regulated call; safety boundary adversarially tested.

AI on FHIR — Clinical NLP & LLM API Integration

Generative AI · HIPAA Guardrails

2025 · Python, Flask, React, TypeScript, Docker, BERT, spaCy, LLMs, FHIR REST APIs · github.com/venkatalakshmiparimala/AI_on_FHIR

- NLP pipeline translating English/Spanish clinical questions into FHIR API calls using LLMs, BERT, and embeddings; **context gathering** from structured medical records; HIPAA guardrails: RBAC for data isolation, output filtering for PHI; ~60% reduction in manual clinical lookup time; containerized with Docker.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering — JNTUA, India — 2013–2017

TECHNICAL SKILLS

Agentic AI & LLMs: Agentic Systems, Agentic Pipelines, Agentic Infrastructure, Multi-Agent Architectures, Context Engineering, Token Optimization, Context Gathering, Tool Selection, Multi-Step Planning, Experimentation, LLM Evaluation Frameworks, LLM-as-Judge, Eval CI/CD, RAG Systems, RAG Pipelines, Hybrid Retrieval, SOTA Model Integration, Generative AI, LangGraph, LangChain, LangGraph StateGraph, Anthropic Claude, OpenAI, Azure OpenAI, Groq, Ollama, Prompt Management, Structured Outputs, Human-in-the-Loop (HITL), Guardrails, Hallucination Detection, Fallback Paths

ML & Fine-Tuning: Fine-Tuning (HuggingFace Transformers + PyTorch), Transfer Learning, BERT, spaCy, scikit-learn, XGBoost, Isolation Forest, Embeddings, Semantic Search, ChromaDB, Pinecone, Weaviate, Elasticsearch, TF-IDF, Vector Search, OCR

Engineering & Infrastructure: Python (9+ yrs), FastAPI, Flask, Django, Java, Spring Boot, TypeScript, Node.js, React, PHP, SQL, PL/SQL, Bash, PostgreSQL, MySQL, Oracle, Redis, MongoDB, SQLite, Snowflake, ETL Pipelines, Data Modeling, Query Optimization

Cloud, DevOps & Observability: AWS (S3, EC2, RDS, Lambda, Glue, SQS, API Gateway, IAM), Azure (OpenAI, Data Factory, Databricks, AKS), GCP (Vertex AI), Docker, Kubernetes, Terraform, GitHub Actions CI/CD, OpenTelemetry, Prometheus, Grafana, RabbitMQ, Kafka, Celery